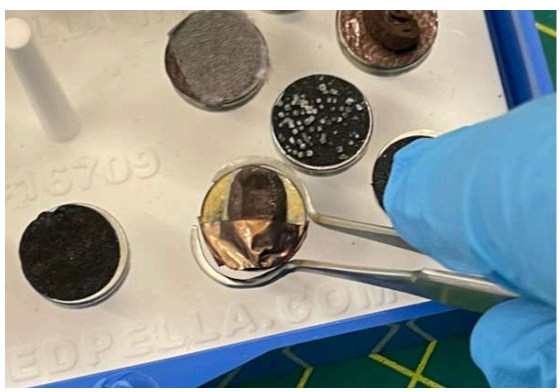
3.000 - Scanning Electron Microscope Station

In this rotation you will look at cross-sections of green and roasted coffee beans using the SEM, and see how roasting changes the structure of the bean. *You will need to save two photos: one of the cavities of a roasted and one of a green bean. You will submit these to Canvas once this rotation is complete.*

Instructions:

1. Put on gloves. These are available on the left central table when you walk in the door to the lab.

2. Use the curved tweezers to pick up the SEM sample stud with the ROASTED coffee bean.
    a. This coffee bean is ion milled to reveal its inner structure. It is also gold coated to make it conductive. This prevents electrons from building up on the sample surface.



3. Spray the bean with compressed air for 1-2 seconds at a distance of 6 inches.

4. Push the sample stud into the black sample holder.
    a. The black sample holder is the default sample holder and tells the machine to pull a strong vacuum. If your sample is nonconductive, you should use the gray sample holder, which pulls a weaker vacuum. There is also a sample holder which freezes your sample, but this requires additional training.

5. Lower the sample until the very top of the bean is in line with the edge of the sample holder by rotating the silver ring at the top of the sample holder.

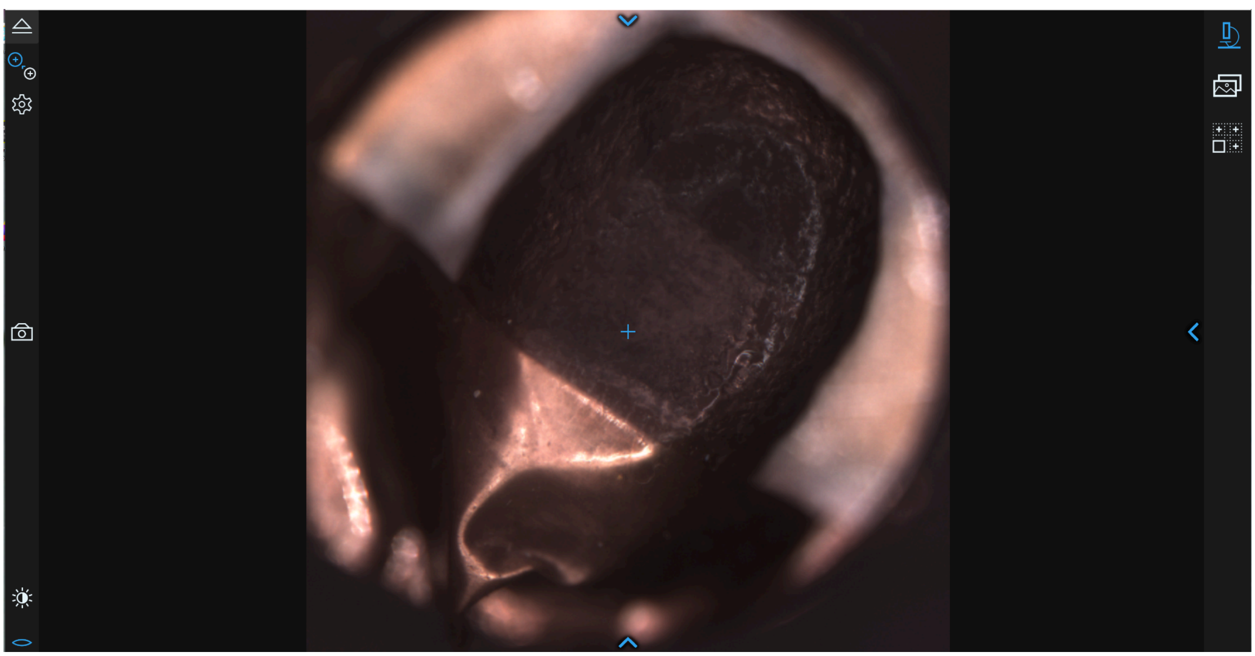
6. Insert the sample holder into the SEM with the chip facing into the machine.

7. Close the SEM chamber by pushing the silver ledge of the door down.
    a. NEVER put wet or dusty samples in the SEM. The strong vacuum will vaporize any water and harm the SEM detector.

8. Log onto the computer using your MIT kerberos.

9. On the computer, click the logo for "Phenom User Interface"
    a. Take your gloves off when using the computer

10. The software should open up onto an optical microscope view of the sample. If this doesn't happen, call over a TA.



Optical microscope view

11. Press the arrow on the right hand side of the screen ( ) to open the panel where you can adjust the focus, contrast and brightness.

12. Once satisfied click, Move to SEM on the top left corner of the screen ( )

13. Now you are in the scanning electron microscope view! Play around with the controls to find interesting views and positions. You will need to submit one or more photos you take in this view. Here are some controls you may need:

Taking a picture: Click the camera icon ( ) on the left side of the screen.

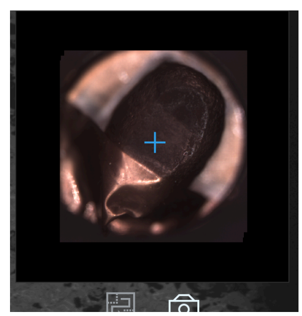
Adjust the quality of your picture: Click the arrow at the top of your screen ( ). Under the sublevel "Acquisition" you can change how much data you average over via “Averaging” or what quality you want your image to be under “Scan Size”. I recommend Averaging: Best and Scan Size: 1920x1080.

Zooming:
Option 1: Using the middle scroll button of your mouse
Option 2: Left click and hold to form a box. The image will zoom to fit the bounds of that box.

Option 3: Click the right arrow ( ) and adjust the knob for Magnification

To move around the image:
Small distances: Click anywhere on the image to move to that spot.

Large distances: Click the right arrow ( ) and click anywhere on the optical microscope photo of your sample to move to that spot.



Change focus:

Option 1: Click the right arrow ( ) and adjust the knob for Focus (best for large changes in focus)

Option 2: Click AutoFocus button ( )
Option 3: Right click with your mouse and move your mouse left or right (this fine tunes focus)

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14. When you’re done, click the button to Unload Sample ( ), and confirm yes when it prompts you to confirm that you’d like to unload your sample.

15. Open the door and remove your sample. Perform the steps to prepare the same in reverse.
    a. Put on gloves.
    b. Pull out the sample holder.
    c. Rotate the silver ring to bring the sample to its max height.
    d. Remove the sample with the curved tweezers.
    e. Return the sample to the sample box.

16. Now, complete the steps above with a **GREEN** coffee bean.

17. To access your image, it will either be in <YOUR_KERB>\PhenomAcquisitions or This PC/Pictures.

18. To save your image, you can email it to yourself, move it to a USB stick or move it to the common Breakerspace Dropbox. Visit https://breakerspace.mit.edu/resources.html to learn how to sign up to the Breakerspace Dropbox